

BE 503: Biosensing and Bioinstrumentation

Course Code	BE 503
Course Name	Biosensing and Bioinstrumentation
L-T-P-C	3-0-2-4
Prerequisites	IC161, IC161P, IC260 or equivalent
Intended for	B. Tech M.Tech Integrated Dual Degree Bioengineering students
Distribution	Core for IDD BE, Elective for other B.Tech students
Approval	44th BoA (Nov 2021); 35th Senate (Jun 2022), in the Second Major in Bioengineering curriculum
Revised	49th BoA (Mar 2023) - revised

Course Contents

- **Measurement**
SI units, systematic and random errors in measurement, expression of uncertainty - accuracy and precision index, propagation of errors, DC potentiometer; bridges for measurement of R, L and C, Q-meter, signal-to-noise ratio, responsivity of a sensor (Transformation of Input-to-output signal). [4 Lectures]
- **Biosignals**
Origin, nature, and types of Biosignals, Principles of sensing physiological parameters from plants and animals, Bioelectric signals and their characteristics. Chemical and electrochemical biosignals. [3 Lectures]
- **Sensors**
Classification of transducers and their characteristics, viz. Voltage sensors, Optical sensors, Displacement/Pressure sensors and accelerometers, Chemical sensors, Acoustic sensors – basic principles, signal conditioning considerations, examples (e.g. biopotential electrodes, pulse oximeter, glucose monitor, hearing aid, etc.); Physical biosensors and associated signal conditioning circuits; Chemical biosensors; Antibody based biosensors, DNA based biosensor, Immunoassays for plant and animal pathogen detection, Enzyme linked immune-sorbent assays (ELISA), bio-luminescent technologies for pathogen detection; Optical sources and detectors: LED, Photo-diode, p-i-n and avalanche photo diode, optical interferometers: applications in metrology; basics of optical sensing and LASER; basics of magnetic sensing. [16 Lectures]
- **Bioinstrumentation**
Biopotential Amplifiers, Noise and artefacts and their management, Electrical Isolation (optical and electrical) and Safety of bio-instruments. Generation, Acquisition, and signal conditioning and analysis of biosignals. Principles of measuring blood pressure, bioamperometric enzyme electrode. [15 Lectures]
- **Analytical bio-techniques**
Principles and applications of UV-Visible-NIR spectroscopy, fluorescence spectroscopy, MR spectroscopy, basics of chromatographic techniques, imaging techniques – principles and applications of microscopy. [4 Lectures]

Laboratory Modules

Design of measurement circuits, ELISA test, extraction of bio-signals, amplification and isolation of bio-signals, phase contrast microscopy, chemoluminescence, fluorescence spectroscopy, MR spectroscopy, spectrophotometry, Pulse oximeter, blood pressure measurement device.

Text Books

1. A.G. Webb, **Principles of Biomedical Instrumentation**, Cambridge University Press, 2018.
2. J. G. Webster, **Medical Instrumentation – Application and Design**, 4th edition, John Wiley and Sons, 2020.

References

1. R.S. Khandpur, **Biomedical Instrumentation – Technology and Applications**, Tata McGrawHill, 2017.
2. CS.C. Mukhopadhyay, A.L. Ekuakille, **Advances in Biomedical Sensing, Measurements, Instrumentation and Systems**, Springer-Verlag, Germany, 2018.
3. J. R. Lakowicz, **Principles of Fluorescence Spectroscopy**, Springer Science & Business Media, 2013.
4. H. H. Willard, L. L. Merritt, J. A. Dean, F. A. Settle **Instrumental Methods of Analysis**, CBS publishers and Distributors, 2015.